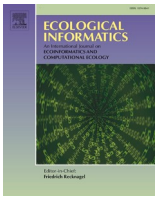
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Improving salmon marine survival models with covariance map indices of sea surface temperature (CMISST) and sea surface height (CMISSH)

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ABSTRACT

Pacific-basin scale atmospheric and oceanographic conditions cascade to regional environmental conditions. In turn, these regional conditions in the nearshore ocean have direct and indirect impacts on coastal ecology and influence everything from primary productivity to top predator abundance and distribution. Quantitatively representing these large-scale conditions with consolidated metrics allows the ocean environment to be represented in a small number of variables, or indices, and can aid in fisheries management and policy decisions. As an example, Pacific salmon face numerous challenges during their migration from freshwater rivers to the ocean and back, and managers must make accurate assessments of their marine survival to inform effective conservation and management strategies. From satellite data and earth systems models, we now have rich data sets with complete spatial and temporal coverage to use for these applications. Here, we create a spatial map of covariance values between salmon and sea surface temperature (or height) and describe a method to calculate a temporal index of similarity to that map. The resulting index, referred to as a Covariance Map Index (CMI), can be used in subsequent predictive models of salmon adult return abundance or survival. Importantly, and unlike standard indices such as PDO, this new metric is tailored for a specific stock of salmon, and can be easily applied to other marine species. By employing this innovative tool, researchers, policymakers, and conservationists can gain deeper insights into marine survival patterns, enabling more informed and effective strategies for the sustainable management of these iconic species.

1. Introduction

Predicting the future is a multifaceted challenge that can draw upon diverse fields such as economics, climate science, and machine learning, employing an array of methodologies. In the realm of managing marine species, we rely on empirical evidence and a blend of qualitative and quantitative models to estimate factors like abundance, distribution, and survival of various species (Cury et al., 2008; McClure et al., 2013; Robinson et al., 2017). These predictions play a critical role in guiding decisions related to human activities and the management of harvesting practices. Predictive models can range from intricate stock assessments and estimations of demographic rates to simpler linear regressions, all aimed at estimating the current or future state of a resource (Walters and Martell, 2004; Ward et al., 2014).

For marine species management, each of these modeling approaches can be aided by accurately representing the ocean environment, though this is made difficult by the complex and dynamic processes influencing most species. Lacking the ability to precisely represent a species' experiences, scientists and managers often simplify the ocean environment using index variables that represent the state of an unmeasured biological or physical condition or process (Harvey et al., 2017). These can take the form of a variable averaged over large spatial scales (e.g., sea surface temperature, sensu Friedland et al., 2014), a variable chosen from a specific location (e.g., buoy data), or a larger set of metrics that have been statistically condensed into a small number of summary metrics and used as an index reflecting integrated conditions (via Dynamic Factor Analysis or similar method, e.g., the community state index described by Hunsicker et al. (2022)). In all cases, the variables do

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not represent all of the complex ecological mechanisms at play, but are assumed to be somewhat correlated with, and therefore reflect, those processes.

Pacific salmon (*Oncorhynchus* spp.) serve as an appropriate test case to improve how we characterize the response of fish to conditions in the ocean for several reasons: i) the ocean environment has a significant impact on salmon population dynamics (Beamish and Mahnken, 2001; Mantua et al., 1997), ii) different species or populations of salmon have unique migration behavior, resulting in varied ocean experiences (Tucker et al., 2011; Teel et al., 2015), and iii) there is an urgent need to quantitatively represent the marine environment in salmon life-cycle models and management efforts (Crozier et al., 2021). Importantly for salmon, which return to their natal spawning location in fresh water, local or regional management bodies often create forecasting models to better prepare for the potential number of adult salmon returning to their system each year. As a result, models of abundance or survival tend to be specific to a population or Evolutionarily Significant Unit (ESU, Waples, 1995). For instance, state agencies use forecast models to estimate the number of salmon at sea and of adult salmon returning the following year; this is essential for setting harvest limits and research permits for allowable take of federally protected species. Similarly, hatchery managers require estimates of returning broodstock to plan procurement, staff hiring, and facility maintenance or expansion.

This diversity in management needs gives rise to a multitude of model types, applications, and data requirements. Many models that predict salmon survival or adult abundance rely on abundance data for younger age classes from a given cohort or brood year to forecast older age classes that return the following year, known as sibling regression (Haeseker et al., 2007). These methods typically prove effective by removing variability associated with early ocean mortality from the assessment and focusing instead on an assumed maturation rate of fish returning earlier from the cohort (Beamish and Mahnken, 2001; Haeseker et al., 2007). In addition, forecast models can incorporate environmental variables, recognizing the substantial influence of environmental drivers on survival, especially during the initial year in the ocean (Pacific Fishery Management Council, 2019). For example, basin-scale indices such as the Pacific Decadal Oscillation (PDO) or the North Pacific Gyre Oscillation (NPGO) have been widely used for qualitative salmon forecasting models along the coast of California, Oregon, and Washington due to their strong correlation with multiple physical and biological ocean processes and salmon responses in this region (e.g., Litzow et al., 2020; Logerwell et al., 2003).

Many models of salmon marine survival or adult return abundance use generalized variables to represent large-scale ocean conditions rather than something specifically tailored for the given species or population, despite the fact that they migrate at different times and to different places, resulting in variable ocean experiences. The diverse behaviors and distributions of salmon species or populations in the ocean render any single basin-scale index ineffective for all groups, as they can have very different production patterns (Hare et al., 1999). Moreover, recent data indicate shifting correlations between indices like the PDO and salmon abundance or survival (Feddern et al., 2025; Litzow et al., 2020), diminishing the predictive ability of many commonly-used indices. Climate-driven ecosystem changes are only exacerbating the gap between reliable and readily-available environmental indices and salmon marine survival. Unfortunately, when local or regional-scale variables have been tested to improve assessment, they similarly have been found to be insignificant contributors to assessment precision and accuracy (Winship et al., 2015).

Hence, our objective was to devise a tool using publicly-available marine datasets to enhance our ability to forecast biological time series and facilitate improved management decisions, recognizing the varied ocean experiences among salmon species or populations. To this end, we generated specific indices of environmental variability as an alternative to generalized indices like the PDO and NPGO. To do this, we chose to use both sea surface temperature (SST) and sea surface height (SSH) data, as they are the base of the PDO and NPGO respectively. Specifically, we used information from a spatial map of covariance between sea surface temperature (SST) or sea surface height (SSH) and adult salmon return abundance. Rather than drawing conclusions from the covariance map visually, we describe a method to quantify a time series depicting similarity to the map. As a product emerging from covariance maps, these new indices are specifically tailored to the response variable and represent a novel method for calculating an environmental covariate for forecasting models. We describe the resulting Covariance Map Index of Sea Surface Temperature (CMISST) (and Sea Surface Height, CMISSH) and explore how its performance varies among response variables, time series lengths, and spatial extents.

2. Methods

2.1. Study area

The nominal study area ranged from 10° to 62° latitude and from 158°E to −114°W longitude, representing most of the North Pacific Ocean. However, the method described here can easily be modified to cover any extent desired. Similarly, the spatial grain can vary - we used sea surface temperature data on a 2° x 2° grid and sea surface height data on a 0.33° x 1° grid, but any resolution can be used.

2.2. Data

We had two requirements of any data source we considered. First, it needed to be freely available to the public so anyone can apply our methods. And second, data needed to be both spatially and temporally explicit. Although spatial extent and resolution could be important, we were not sure what the best choices might be, or how they could influence results. Temporal coverage should minimally cover the range of years for which the response variable exists. Monthly or seasonal data are ideal, but even annual data could work if the correlation with the response variable is sufficiently strong.

2.2.1. Salmon counts

Ideally, when modeling SST or SSH effects on salmonids, we would obtain marine survival rates for each species or population of interest. Unfortunately, for many salmonids, we lack the smolt and adult counts required to estimate these rates, and must rely on spawner counts, redd counts, or other indices of abundance. For simplicity in demonstrating our approach, and to better match the situation numerous managers face, we used the number of adult salmon and steelhead observed returning to a particular location as our response variables. Specifically, we used counts of adults at Bonneville Dam, the lowest dam on the Columbia River. However, were direct measures of marine survival to become available, and for populations where it might already be available, the approach described here would be equally appropriate and would avoid potential impacts on the response variable from unrelated sources, such as freshwater survival and marine harvest. We downloaded annual counts for spring Chinook salmon (*O. tshawytscha*), fall Chinook salmon, and steelhead (*O. mykiss*) from Columbia Basin Research's Data Access in Real Time (DART) (https://www.cbr.washington.edu/dart/query/adult_annual_sum). Although each of these groups can be divided into smaller populations or units that might be more appropriate for management decisions, we opted to keep them aggregated for demonstrating the utility of our approach, and hereafter referred to them with the generic term 'stocks'.

Interannual adult salmonid counts at Bonneville Dam were slightly right skewed and were therefore natural-log transformed prior to analysis. To match the temporal extent of available environmental data, we used adult salmon counts from 1982 to 2024. Maturation and age at return can vary for each of the three response variables. To align salmon counts to the year of ocean entry, a period generally understood to be a strong driver of marine survival (Beamish and Mahnken, 2001), we

lagged the salmon counts by the number of years required to convert adult return years 1982–2024 to smolt years 1980–2022; the specific value of the assumed lag was explored (see Sensitivity Analysis below).

2.2.2. Oceanographic data

For SST, we used the Extended Reconstructed Sea Surface Temperature (ERSST) version 5 dataset (Huang et al., 2017), provided by the NOAA Physical Science Laboratory (PSL), Boulder, Colorado, USA (https://psl.noaa.gov). This is a global monthly sea surface temperature dataset derived from the International Comprehensive Ocean-Atmosphere Dataset. ERSST is on a 2° x 2° grid from January 1854 to 2024 and includes anomalies computed with respect to a 1991–2020 monthly climatology. Monthly values of SST tend to have high autocorrelation (Mass et al., 2022). We therefore averaged the SST anomaly data into 3-month bins to get seasonal anomaly values. Winter values encompass January, February, and March. Spring, summer, and autumn values are the subsequent months, in 3-month increments.

We obtained SSH data from the National Centers for Environmental Prediction (NCEP), Global Ocean Data Assimilation System (GODAS; Behringer et al., 1998). This dataset contains monthly SSH anomaly data from January 1980 to present (relative to the climatological period of 1991 to 2020). Data are gridded (0.333° latitude x 1.0° longitude) between 74.5S and 64.5 N latitude and 0.5E and 359.5E longitude). NCEP GODAS data were served online by the NOAA PSL, Boulder, Colorado, USA (https://psl.noaa.gov). Similar to SST, we took 3-month averages of SSH data to represent seasonal anomalies. For both SST and SSH data, we retained subsets of the data to represent smolt outmigration years (1980–2022).

2.3. Covariance maps

Prior to analysis, we scaled each dataset by subtracting its mean and dividing by its standard deviation, producing *Z*-scored data. We quantified the relationship between SST (or SSH) and salmon counts in each grid cell by calculating their temporal covariance. Using scaled data, the covariance is mathematically identical to a linear regression coefficient, had we run a linear regression in each cell.

We can visualize the calculated covariances by plotting them in space (Schroeder et al. 2009), resulting in a spatially-explicit covariance map for any pair of environmental data and salmon response (Fig. 1). An important attribute of these maps is that, by definition, salmon returns have been historically high during periods when mapped SST (or SSH) anomalies exhibited a similar spatial pattern as the covariance map. That is, the best salmon returns occurred during periods when locations in the map with strong positive covariances had high SST (or SSH) anomalies and locations with negative covariances had low SST (or SSH) anomalies. We can therefore think of these covariance maps as an optimal spatial pattern of SST (or SSH) anomalies, and quantify how similar any given year is to this optimal pattern. By doing so, we can essentially mimic a common and successful nearest-neighbor approach to forecasting (e.g., Tajmouati et al., 2024).

Although the covariance map displays the spatial pattern of how SST (or SSH) covaries with a given response variable, in this case salmon, the spatial patterns should not be interpreted as indicating where salmon migrate. Yet, they can be informative in a comparative sense. For example, comparing these patterns across responses (species or populations), or to spatial patterns from other predictive environmental metrics such as the PDO or NPGO, can provide clues as to how ocean dynamics might impact the various responses.

2.4. Covariance map index (CMI)

We created the Covariance Map Index (CMI) as follows. To use indices of sea surface temperature (CMISST) and sea surface height (CMISSH) in a forecasting scenario, we needed to convert a given SST (or SSH) dataset into a time series. By vectorizing and regressing the SST (or

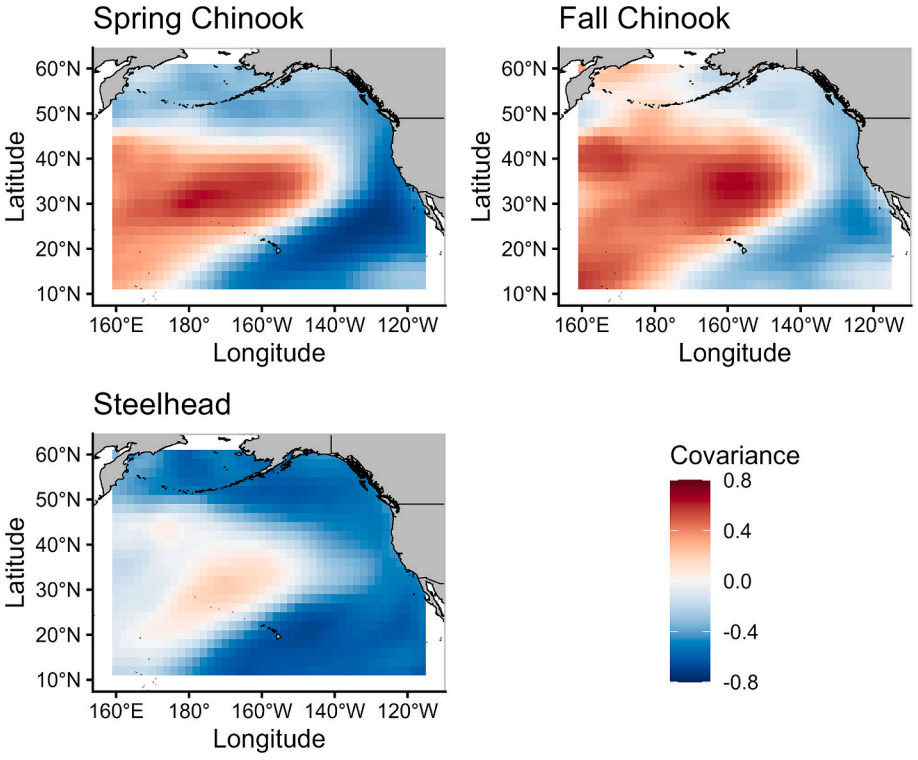


Fig. 1. Covariance Map for adult counts at Bonneville Dam from 1982 to 2024 (ocean entry years 1980–2022). Each grid cell represents the covariance between adult counts at the dam and sea surface temperature in that location during spring (April–June) lagged by 2 years. Collectively, these maps represent the benchmark spatial pattern of how salmon returns relate to SST. Years with warmer than average SST in the central North Pacific and cooler than average SST along the North American coast typically result in the highest salmon returns.

SSH) dataset onto the vectorized covariance map, the estimated regression coefficient becomes an index that describes how similar any given spatial pattern in SST (or SSH) is to the benchmark pattern exhibited by the covariance map (see Fig. 1):

$$Y_{ti} \sim \beta_t \, Cov_i + e_{ti}$$

Where Y_{ti} is a vector of either SST or SSH at time t and cell i, β_t is the regression coefficient at time t, Cov_i is the covariance in cell i, and e_{ti} represents the normal error for each time t and cell i. Periods when SST (or SSH) have high spatial similarity to the benchmark pattern in the covariance map will have a large positive regression coefficient (β_t) and periods that are opposite of the benchmark spatial pattern will have large negative coefficients. CMI is simply the time series of β_t values. For the current application, the β_t terms are independent, although this assumption can be relaxed, allowing for more temporal autocorrelation in the resulting index.

2.5. Evaluating CMI performance

Because we used information from the response variable to create our CMIs, using measures of model fit may bias our estimation of model performance. We therefore made all model performance comparisons using leave-one-out cross validation (LOOCV), where we removed one year from the dataset, recreated the CMIs from the remaining years, then used that modified index to predict the response variable from the removed year. As a result, all models, regardless of which covariate was used, were evaluated using out-of-sample data.

Given that the goal of creating CMI was to create a stock-specific covariate for models of salmon adult abundance, we compared the fit of a regression model using CMI as a covariate to models that instead used other available covariates. We compared the predictive ability of CMI (CMISST and CMISSH) to that of the PDO (Mantua et al., 1997), NPGO (DiLorenzo et al. 2008), Ocean Niño Index (ONI; https://origin.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php), and SSTarc, an index of SST anomalies for the region defined by Johnstone and Mantua (2014). Data sources for these indices are listed in 'Data Availability.' As with CMI, we binned each index into seasonal means.

These methods are intended to improve management effectiveness when using forecast models, where even small improvements in predictive ability could save money or more effectively aid in recovery. To directly compare and contrast CMIs and other environmental covariates, we calculated multiple performance metrics across all available years, including the root mean squared error (RMSE), mean absolute error (MAE), and Kling-Gupta Efficiency (KGE). We also computed correlations among indices and between salmon counts and indices.

2.6. Sensitivity analyses

Many Chinook and steelhead stocks leave the Columbia River and spend an average of two to three years in the ocean. To estimate the impact of assuming a 2-year lag, we sequentially ran the model with lags ranging from 0 to 4 years and calculated the effect on model performance.

An important attribute of this new index is that it was created with the assumption of a stationary linear relationship between patterns in SST (or SSH) and salmon returns. Because correlations can shift over time (Litzow et al., 2020), we evaluated the impact on model performance of using a variable period of data to create the CMI. We started with the full time series and sequentially removed early data to recreate the index with ever shorter time series. That is, we created the nominal CMISST using salmon return data from 1982 through 2024. We then created the index using 1983–2024, then 1984–2024, etc., and continued until we created an index using only 2015–2024. If non-stationarity exists in the relationship between spatial patterns in SST and salmon returns, we would expect to see performance to differ across these subsets of the data. Because it would be inappropriate to compare the RMSE across time series with different lengths, we performed this LOOCV on the most recent 10-year time period (the length of the smallest period evaluated) within each dataset. For this evaluation, we set the lag for each stock to that with the lowest RMSE. For simplicity, we only performed sensitivity analyses on CMISST (not CMISSH).

We applied similar logic to the spatial extent as we did to the temporal extent of the SST data. Using LOOCV, we tested model performance when the southern extent of the spatial grid started at 10° intervals between 40°S and 50°N latitude to quantify the impact of variable spatial extents on model performance. We also tested the western edge of the spatial extent, allowing it to range from 100°E to 220°E longitude. Because many salmon stocks migrate along the northwest coast of North America, we retained the eastern and northern edges of the spatial grid at 114°W and 62°N, respectively, to ensure their migration pathway was included in the spatial extent. For spatial sensitivity analyses, we ran the LOOCV across all available years (1982–2024) and set the lag for each stock to that with the lowest RMSE.

3. Results

3.1. Stock-specific index performance

The new stock-specific map-based indices CMISST and CMISSH had strong correlations with adult salmonid returns (Fig. 2, Fig. S1) and generally showed as good or better predictive ability than even the best of the four commonly-used indices of ocean conditions that we tested (Fig. 3, Fig. 4). Although performance varied by species, lag and season, a CMI almost always had a lower RMSE than any other index for a given season and lag (Fig. 3). For spring Chinook salmon, the best index as determined by lowest RMSE was CMISST in spring with a 2-year lag, the best index for Fall Chinook salmon was CMISSH in autumn with a 3-year lag, and the best index for steelhead was CMISST in autumn with a 3-year lag. Other performance metrics (MAE, KGE) showed similar patterns (Tables S1–3).

Fall Chinook showed the biggest benefit of using a CMI relative to other indices (Fig. 4). Percent improvements in RMSE between the best CMI and the best non-CMI index was 7 % for spring Chinook salmon and steelhead and 13 % for fall Chinook salmon (Tables S1–3). For fall Chinook salmon, CMISSH outperformed all other metrics, including CMISST, in all seasons whereas NPGO and SSTarc performed poorly in all seasons. PDO performed least well for steelhead and ONI typically did not perform well for any stock (Fig. 3).

Many of the example datasets we explored resulted in a spatial covariance pattern similar to the PDO, with low values along the eastern Pacific coast and higher values in the Central North Pacific (Fig. 1). However, the magnitude of the covariance values and the relative influence of positive and negative covariance regions differed among salmon data sets. For spring Chinook salmon and steelhead, there were areas off the western coast of North America with strong negative covariances, whereas fall Chinook showed few areas with negative covariances, but strong positive covariances in the central and western North Pacific. These differences highlight the varied relationships between salmon stocks and basin-scale environmental patterns and the resulting need for stock-specific indices.

3.2. Sensitivity analyses

When we evaluated the temporal lag between ocean conditions and adult returns, we found that a 2-year lag was almost always best for spring Chinook salmon, regardless of the season and index used (Fig. 3). For fall Chinook salmon, 2 and 3-year lags usually performed equally well, though a 3-year lag performed best in autumn. For steelhead, a 2-year lag resulted in the best predictive ability in winter and spring but a three-year lag was better in summer and autumn, with some exceptions. Lags of 0, 1 and 4 never performed better than 2- or 3-year lags. For all

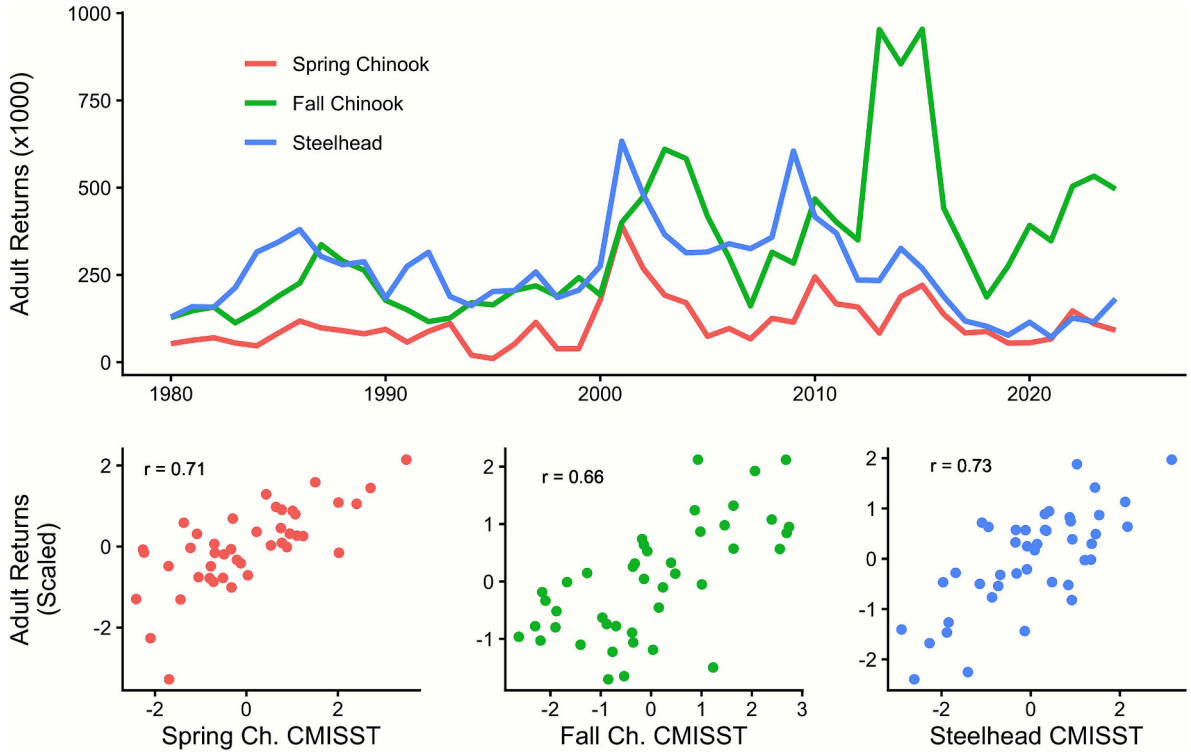


Fig. 2. Time series of adult counts at Bonneville Dam (1982–2024, top) and the relationship between those counts and the CMISST generated from spring (Apr-Jun) SST data for each stock (bottom), assuming a 2-year lag between ocean entry and adult returns.

other model comparisons, we used the season and lag with the lowest RMSE when matching adult return data to the ocean data.

Interestingly, LOOCV model performance was dependent on the temporal (Fig. 5) and spatial (Fig. 6) extents used to create the CMI. For spring Chinook salmon, predictive ability was highest when using a relatively long time series, starting at least from out-migration year 1990. In contrast, the steelhead model performed best with a relatively short time series, starting between 2005 and 2010. The fall Chinook salmon model did not seem to be as sensitive to time series length. For all three groups, a time series shorter than about 15 years resulted in poor predictive ability.

We did not explore all combinations of starting latitude and starting longitude, but rather looked at deviations from the nominal values for each metric individually. Sensitivity to starting latitude seemed to be relatively low for all stocks until values exceeded 20° N, when model performance degraded. The steelhead models were only modestly sensitive to starting longitude. In contrast, the fall Chinook salmon models improved when including more longitudinal data, as far west as 120°E. Oddly, there is a very strong positive relationship between fall Chinook returns to the Columbia River and SST in the Philippine Sea, just northwest of the El Niño Southern Oscillation Region 4 (https://www.ncei.noaa.gov/access/monitoring/enso/sst).

4. Discussion

The development of predictive tools for marine species, including Pacific salmon, is critical given the complexities of oceanic processes and their influence on population dynamics. Our results demonstrate that stock-specific Covariance Map Indices (CMI), such as the Covariance Map Index of Sea Surface Temperature (CMISST) and Sea Surface Height (CMISSH), can improve predictive accuracy by tailoring relationships with environmental variability to individual salmon stocks. This approach directly addresses the limitations of generalized indices, which may not reflect the unique ecological experiences of different stocks. Notably, CMISST and CMISSH showed stronger correlations with adult returns for spring Chinook salmon, fall Chinook salmon, and steelhead than other indices, further suggesting that CMIs can enhance the forecasting models used in management and conservation efforts.

While basin-scale indices have historically provided useful correlations, their usefulness is mixed and appears to be waning due to climate-driven changes in ecosystem dynamics (Feddern et al., 2025; Litzow et al., 2020). For example, the PDO, which has been widely used by hatchery and harvest managers, consistently underperformed, especially for steelhead populations, revealing a significant gap in forecasting accuracy. As a result, harvest management of Columbia River steelhead is based on the average returns during recent years and sibling regression, rather than an environmental covariate-based forecasting model, due to the poor performance of existing metrics (U.S. V. Oregon Technical Advisory Committee, 2017).

4.1. Stock-specific indices consistently showed high predictive skill

CMI is the only index of those tested capable of specializing by stock. By evaluating the performance of the stock-specific indices CMISST and CMISSH against traditional indices like the PDO and NPGO, we highlighted the value of new quantitative approaches. CMISST and CMISSH consistently exhibited high predictive skill, often offering moderate to substantial improvements over established, generalized indices for representing the diverse oceanic experiences of different salmon stocks.

This highlights the limitations of existing indices when applied broadly across stocks. Typical indices are static and represent large-scale oceanographic patterns (e.g., Mantua et al., 1997), often lacking the spatial and temporal specificity needed to reflect the migratory and ecological nuances of particular salmon stocks (Wells et al., 2020). As a result, practitioners must often search across numerous indices to find one that correlates well with their particular stock of interest, creating a patchwork of environmental covariates. As might thus be expected, the particular non-CMI index most highly correlated with the best CMI differed among stocks (Fig. S1). This finding highlights that with CMIs, a single approach with consistently high predictive ability across stocks

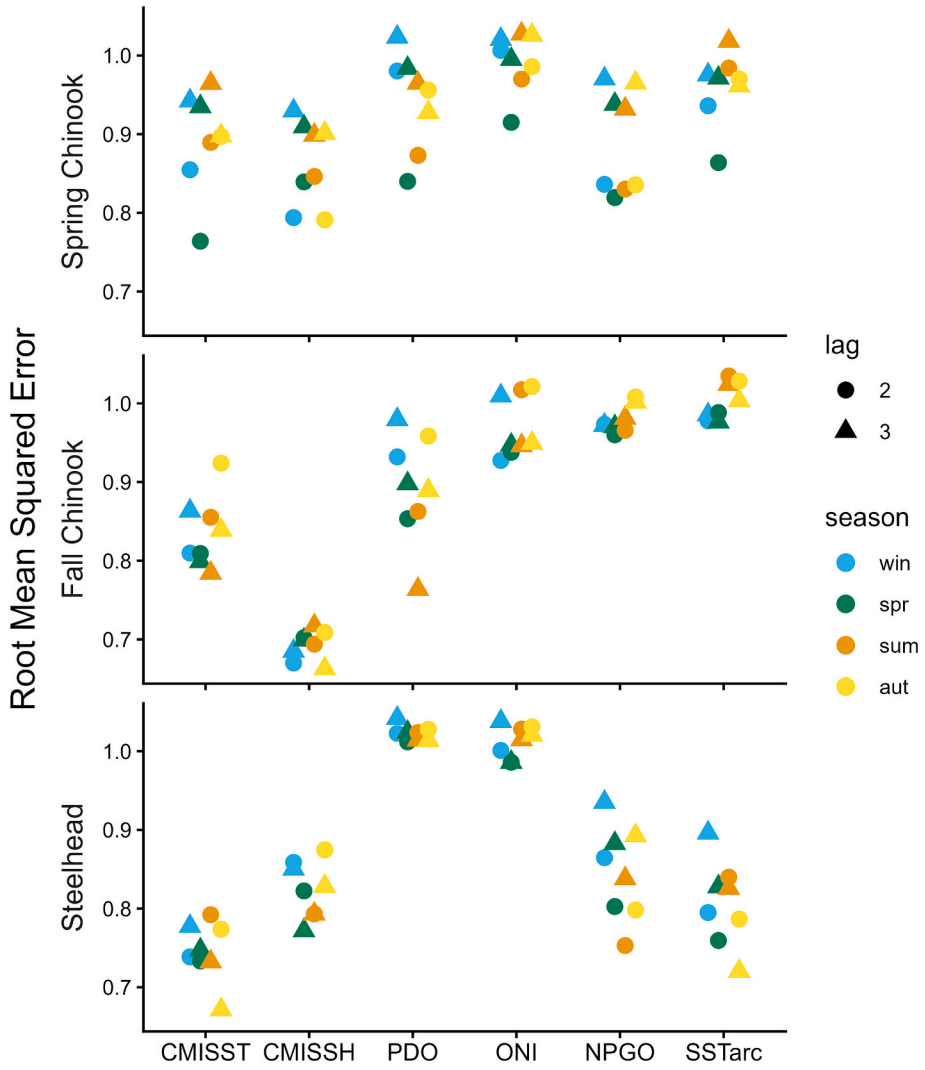


Fig. 3. Root Mean Squared Errors (RMSE) for a leave-one-out cross validation of individual indices (CMISST, CMISSH, PDO, ONI, NPGO, or SSTarc) linearly regressed on standardized adult salmon counts from 1982 to 2024. Panels show scores for spring Chinook salmon (top), fall Chinook salmon (middle), and steelhead (bottom) by index. Colors represent seasons and symbol shapes represent lags of 2 or 3 years between smolt outmigration and adult returns. Lower RMSE values indicate better predictions. For each stock, the index, season, and lag with the lowest RMSE was used in subsequent analysis and interpretation.

can obviate the need to exhaustively search across multiple disparate metrics.

Flexible, stock-specific indices like CMISST and CMISSH better reflect the complex and shifting environmental drivers influencing marine survival and abundance compared to traditional indices. Our stock-specific indices allow tailored insights that better align large-scale marine patterns with the unobserved experiences of different salmon stocks, ultimately leading to more accurate predictions.

4.2. *Insights from sensitivity analysis and caveats*

Our sensitivity analysis of the CMIs revealed notable differences in how analytical assumptions impacted predictive performance across different salmon stocks and seasons. This variability highlights the importance of carefully considering methodological choices and data sources in building predictive models. But it also highlights the flexibility of CMIs relative to traditional indices: their predictive performance can be optimized by fine-tuning options such as starting latitude, longitude, and time series length.

Sensitivity analysis underscored the significance of age composition of salmon stocks and their lagged responses to ocean conditions. The majority of fish spend two years in the ocean before returning to spawn (Beamish, 2018), and our use of a constant 2- or 3-year lag was an oversimplification of variability in ocean residence time. Yet, when creating the CMI, this assumption allowed us to temporally match ocean conditions with salmon returns without knowing the full age composition of adults, and still resulted in high prediction skill. Predictive ability sometimes shifted across seasons, perhaps reflecting migration by salmon further from their natal rivers over time. For example, CMISST for steelhead with a 3-year lag improved across seasons, whereas a 2-year lag worsened across seasons. This suggests a complex interplay of pre-conditioning processes and pre-spawning dynamics that merit further exploration. A method we did not employ, but many forecasters do, is to model age classes separately and aggregate the forecasts to estimate total returns in a given year. For this, a CMI could be created for each time lag and predictions could be weighted by the estimated age composition of the stock.

The choice of time series length plays a critical role in model sensitivity. Based on our results, we suggest that time series should be at least 15 years long to produce robust results. A long time series improved predictive ability for spring Chinook salmon, but interestingly, fall Chinook was not overly sensitive to time series length. In contrast, a relatively short time series was best for steelhead, perhaps due to a nonstationary relationship between ocean conditions and steelhead

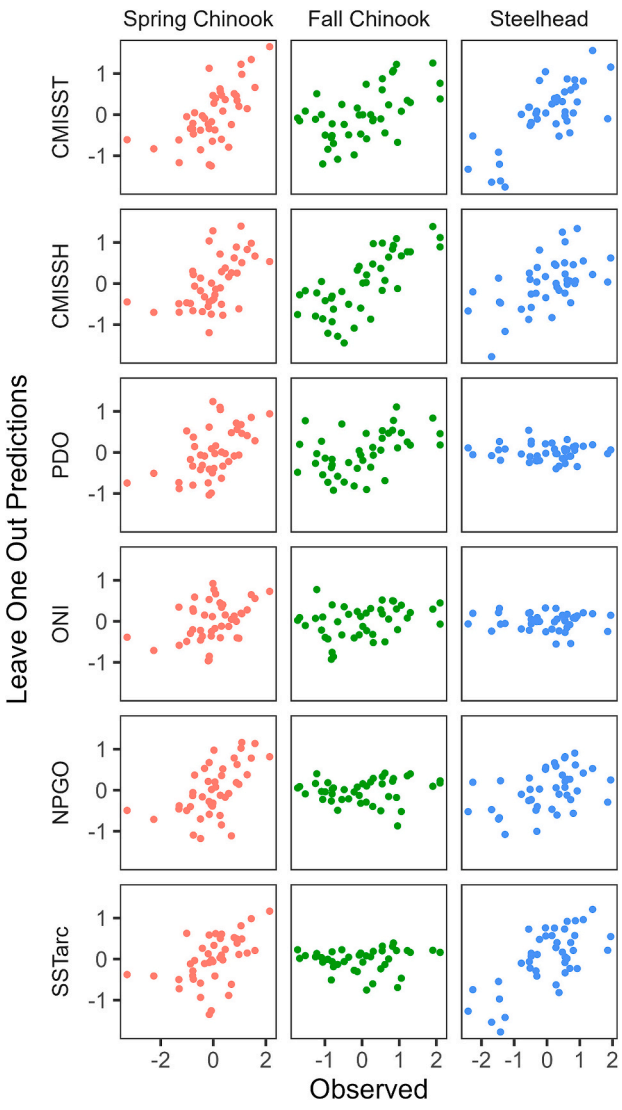


Fig. 4. Standardized values of observed salmon abundance at Bonneville Dam (x-axis) versus predictions from leave a one out cross-validation analysis (y-axis) for each index (rows: CMISST, CMISSH, PDO, ONI, NPGO, and SSTarc) for spring Chinook salmon (left column), fall Chinook salmon (middle column), and steelhead (right column). Colors match those in Fig. 2 to facilitate interpretation.

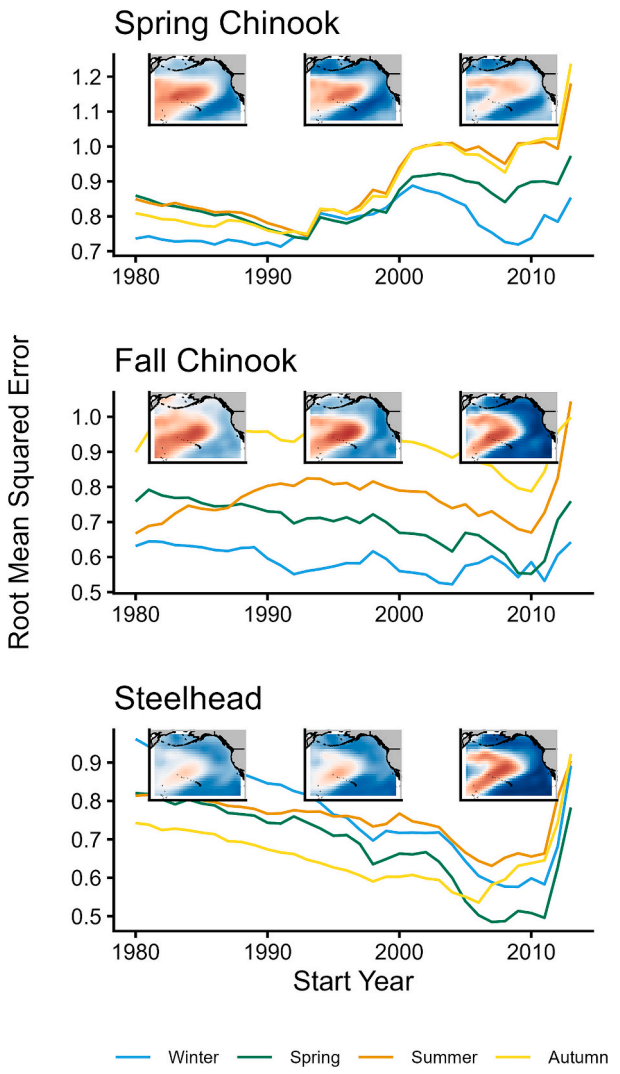


Fig. 5. Root Mean Squared Errors (RMSE) of predictions from a leave-one-out cross validation analysis (y-axis) as a function of the starting year in the time series (x-axis) and by season (colored lines) for spring Chinook salmon (top panel), fall Chinook salmon (middle panel), and steelhead (lower panel). The uptick at the end of many time series illustrates the effect of overfitting when using a very short time series. Inset maps depict how the covariance map changes with different time series lengths (1980–2022, 1995–2022, and 2010–2022); deeper-colored patterns are associated with a lower RMSE.

response. Evidence for similar non-stationarity exists in relationships between traditional ocean indices and salmon returns (e.g., Feddern et al., 2025; Litzow et al., 2018), and quantitative tools are available to handle this - a point to which we return below.

The spatial extent over which we summarized CMI also influenced the predictive ability of the indices. By design, CMI uses large-scale spatial patterns to identify correlative relationships. For fall Chinook in particular, predictive ability improved with increased latitudinal and longitudinal extents. In contrast, spring Chinook and steelhead showed only modest improvements by expanding extent. Ultimately, the choice of spatial extent should be made based on evaluations for any particular response. However, we point out that as spatial extent is reduced, CMI will be less based on spatial patterns and more related to mean SST (or SSH) in the extent used, potentially altering the interpretation of the metric.

Model results may be influenced by the metric we used to represent salmonid responses: we used aggregated counts of adult salmon returning to rivers because a direct measure of marine survival was not readily available for each of these stocks. Hatchery release numbers for Columbia River salmon are relatively stable from year to year (DeHart, 2024), suggesting that adult counts should be correlated with marine survival, but survival of smolts in freshwater can vary among years, potentially degrading the predictive ability of CMIs. Direct measurements of marine survival rates would provide a more appropriate response variable for analysis, were they to become readily available. In addition, marine and estuarine harvest could decouple adult counts from true marine survival. Despite these potential biases, if marine survival estimates were available, we have no reason to think that CMIs would perform less well. Finally, SST and SSH datasets, while widely used, are not immune to measurement errors or biases inherent to their collection and processing, and could impact the calculation of CMIs.

4.3. *Broader ecological insights from the covariance maps*

Covariance maps proved to be valuable tools in our analysis, revealing complex spatial relationships between sea surface temperature or sea surface height and stock-specific adult salmonid returns. By

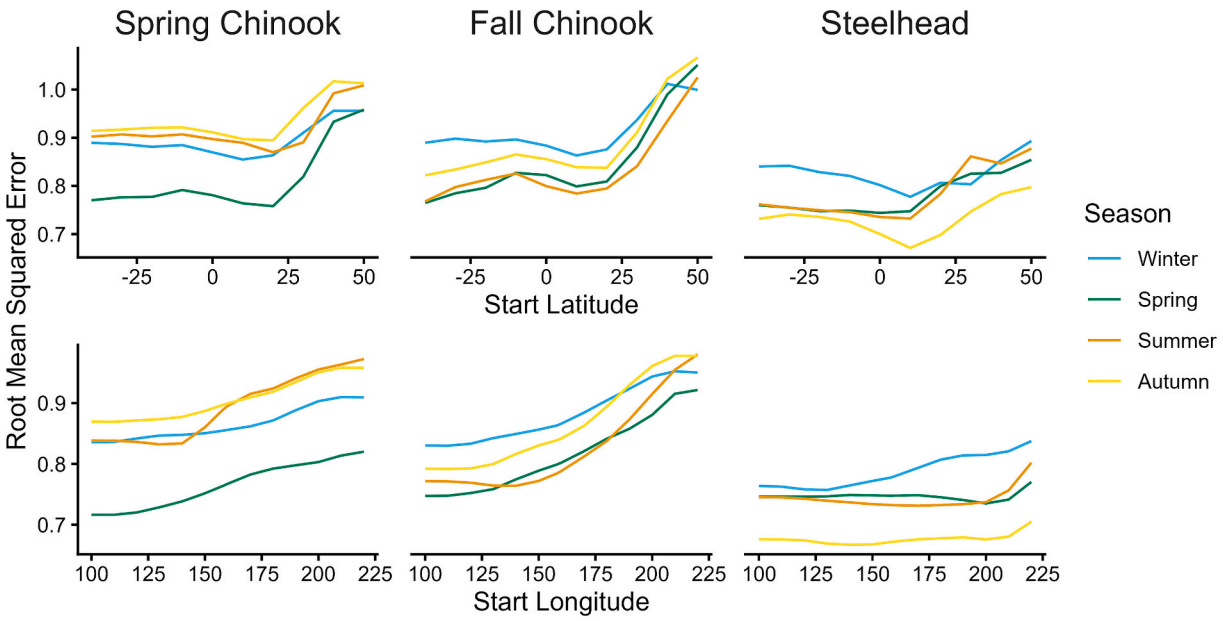


Fig. 6. Root Mean Squared Errors (RMSE) of predictions from a leave-one-out cross validation analysis (y-axis) as a function of the southern (top row) or western (bottom row) boundary of the spatial extent (x-axis) and by season (colored lines) for spring Chinook salmon (left), fall Chinook salmon (middle) and steelhead (right).

capturing correlations between environmental variability and biological outcomes, these maps allowed us to visualize the scale and scope of how different stocks correlate to changes in ocean conditions. The resulting variability in covariance map patterns across the three salmon stocks we studied underscores the importance of accounting for the impact of variability in ecological experiences (partially due to differences in behaviors and migration routes; Larson et al., 2013) when assembling predictive models. For instance, spring Chinook salmon tend to migrate quickly to offshore northern waters, including Alaska and the Bering Sea (Quinn, 2018), while fall Chinook salmon often remain closer to the coast of Oregon and Washington for extended periods (Buckner et al., 2023; Riddell et al., 2018; Weitkamp, 2010). Steelhead exhibit a markedly different migration behavior, quickly venturing into the Central North Pacific (Hayes et al., 2011; Van Doornik et al., 2019). Recent analysis of winter steelhead populations on the Washington coast showed that survival of immature and mature steelhead was negatively correlated with summer SSTs in the western Gulf of Alaska and south of the Aleutian Islands (Ohlberger et al., 2025, also see Fig. 1). These distinct migration behaviors lead to unique ocean experiences for each stock, ultimately contributing to the observed differences in covariance patterns.

Behavioral differences, such as depth preferences, further influence these relationships. Steelhead are surface-oriented, exposing them to highly variable conditions at or near the ocean's surface (Hayes et al., 2011; Light et al., 1989), whereas Chinook salmon's broader depth range (Freshwater et al., 2024; Keefer et al., 2019; Smith et al., 2015; Smith et al., 2024) may lead to interactions with cooler, more stable conditions. This vertical habitat preference likely contributes to differences in the covariance map values observed between the stocks. While the highest covariance values are not necessarily co-located with areas through which the stocks migrate, the varying migration strategies and depth use highlight divergent responses to large-scale ocean dynamics, offering insights into the oceanic influences on each stock's survival.

Spatial patterns in covariance maps also generate hypotheses about broader ecological interactions. For example, contrasting the CMISST covariance patterns for rockfish recruitment (*Sebastes* spp.; data not shown), which differ significantly from Chinook salmon, suggests that favorable ocean conditions for one species may not align with those benefiting another, highlighting the complex ecological dynamics driving various species.

We stress that CMI is not necessarily an index of direct physiological effects (of temperature, for example), as many of the locations used to calculate CMI are outside the spatial extent of most salmon migrations. Rather, it represents associations with broader environmental and ecological processes, including potential shifts in predator and prey distributions. Prey availability can vary due to temperature-related changes in the productivity of lower tropic levels. Predation pressure can vary due to distributional shifts or increased energy demand by predators. Direct effects of temperature on metabolic processes and growth could also lead to altered species interactions, such as increased risk taking by salmon and thus susceptibility to predation. Direct temperature effects and prey availability may also interact; for example, sensitivity to elevated temperatures may be higher when prey is limiting due to higher metabolic stress. Notably, covariance maps extending southward towards the equator often improved model performance. This finding hints at the indirect effects of temperature-driven changes, such as predator distributions (e.g., Pacific hake, Malick et al., 2023) or teleconnections from equatorial regions influencing productivity and survival further north. Broader-scale processes like El Niño can influence bottom-up ecosystem dynamics near the Columbia River (Frischknecht et al., 2017), further demonstrating the utility of covariance maps in linking local conditions to wider oceanic influences.

The strong performance of SSTarc for steelhead also merits attention, as the 'arc' pattern was estimated by Johnstone and Mantua (2014) independent of salmon dynamics. This suggests that the predictive ability of indices like CMISST may be rooted in their capacity to capture strong basin-scale dynamics. Ultimately, these spatial patterns provide a means to identify and compare stock-specific responses, allowing a better understanding of the complex interactions between salmon and their changing ocean environments.

4.4. Management implications

The development and use of stock-specific indices such as CMI present opportunities to enhance the precision and effectiveness of management strategies for salmon and other marine species. Unlike more generalized indices, managers can optimize a stock-specific index for any salmonid response variable, leading to better information to support

management decisions. For example, hatchery managers can use a CMI to better anticipate the number of returning salmon to a particular hatchery and plan operations accordingly, while harvest managers can fine-tune catch limits based on more accurate marine survival forecasts.

The broader applicability of stock-specific indices also offers potential for improving conservation outcomes. By tailoring indices to individual stocks, managers can better ensure compliance with conservation goals, mitigate predation and overharvest, and enhance efforts to recover endangered populations. Additionally, this precision can contribute to meeting treaty obligations that require careful management of shared or transboundary stocks.

However, it is important to note that this approach may not be universally applicable. Stocks with minimal ocean residence or with distributions with relatively uniform ocean conditions may not have strong covariances with SST or SSH. Smaller populations with high interannual variability may exhibit less predictable covariance with SST, potentially undermining the effectiveness of CMI. Our initial exploration of this method with non-salmonid species, such as recruitment of rockfish and sablefish (*Anoplopoma fimbria*) to the juvenile life stage yielded mixed results, suggesting that the approach may be best suited for short-lived species with relatively well-defined life cycles. Long-lived non-anadromous species often exhibit high interannual variability in cohort recruitment due to complex ecological dynamics that could be far-removed from physical variables such as SST.

Acknowledging the challenges posed by climate change and non-stationary environmental conditions, it is clear that effective management must incorporate both immediate, actionable insights and a long-term commitment to better understanding complex ecological processes. CMI and similar tools offer a pragmatic means of addressing current management challenges, while ongoing research can deepen our grasp of the mechanisms driving stock-specific responses. Together, these approaches enable hatchery managers, harvest managers, conservation planners, and policymakers to more accurately predict and respond to fluctuating marine survival rates and abundance, facilitating targeted conservation actions, better planning of hatchery operations, and effective use of harvest limits.

4.5. Conclusion and future directions

The development and application of CMI marks a critical step forward in the management and conservation of salmonids, enabling more informed decisions, mitigating uncertainty and promoting the resilience of marine populations. Importantly, while CMIs outperformed many traditional indices, it is not a panacea. It should be regarded as a complementary tool within a broader management toolkit, capable of addressing immediate needs while informing adaptive strategies over time.

Future research should aim to address key limitations, including the challenge of non-stationarity in the relationship between environmental indices and salmon returns. Climate change and shifting ocean conditions can alter the predictive value of established indices, potentially degrading model accuracy. To address these complexities, techniques such as Dynamic Linear Models could be explored, allowing regression coefficients in each cell to vary over time and better capturing nonlinear, shifting relationships between sea surface temperature and salmon survival. Automating the exploration of model parameters as new data become available could further enhance predictive accuracy, providing a dynamic and flexible approach to ongoing monitoring and management. Traditional indices have been used as covariates in increasingly advanced fisheries forecasting techniques such as machine learning approaches; CMIs could be substituted for traditional indices in those same approaches, likely leading to additional gains in performance.

Refining predictive models also requires a nuanced treatment of age composition in salmon populations. While we restricted our analysis to an assumption of constant ocean residence (using either a 2-year or 3-year lag), future models could incorporate a range of time lags, weighted according to the age composition of returning adults each year. This refinement would provide a more accurate representation of population dynamics and potentially improve the ability to forecast returns of all age classes, addressing a common limitation of sibling regression models that cannot forecast the youngest age class.

In conclusion, stock-specific indices derived from publicly available data have demonstrated strong predictive potential relative to commonly used indices of ocean conditions. Although these indices may not reflect all mechanistic drivers of salmon marine abundance or survival, they offer an important means of making informed, immediate management decisions to support conservation and recovery efforts for salmon and other managed marine resources while also meeting fishing demand and Treaty obligations. By continuing to refine these tools and exploring advanced techniques, managers and scientists can better balance the dual imperatives of species conservation and sustainable fisheries management to ensure the long-term viability of salmon and other marine species.

Author contribution

BJB and AHF conceived the ideas and designed methodology; BJB analysed the data, produced visuals, and led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

CRediT authorship contribution statement

Brian J. Burke: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Aimee H. Fullerton:** Writing – review & editing, Validation, Conceptualization. **Brian K. Wells:** Writing – review & editing, Validation. **Jan Ohlberger:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecoinf.2025.103575.

Data availability

All code was written in R (R Core Team, 2021) and is available to download at https://github.com/noaa-nwfsc/CMISST.git. All necessary data to run the models and recreate the figures for this manuscript are also available in this repository. To make this method easy to use and widely accessible, we created an R Shiny (Chang et al., 2023) visual interface (https://connect.fisheries.noaa.gov/CMISST/). By default, the index is created with SST and counts of adult spring Chinook to Bonneville Dam on the Columbia River (as shown in Fig. 1), but users can select returns of adult fall Chinook salmon or steelhead. In addition, users can upload any time series of interest and the app will create the index, optionally run LOOCV, and plot results. PDO data were obtained from https://oceanview.pfeg.noaa.gov/erddap/tabledap/cciea_OC_PDO.csv. NPGO data were obtained from https://oceanview.pfeg.noaa.gov/erddap/tabledap/cciea_OC_NPGO.csv. ONI data were obtained

from https://oceanview.pfeg.noaa.gov/erddap/tabledap/cciea_OC_ONI.csv. SSTarc data were obtained from https://www.ncei.noaa.gov/pub/data/cmb/ersst/v5/netcdf/ and subset to areas inside the polygon (arc) defined by Johnstone and Mantua, 2014.

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